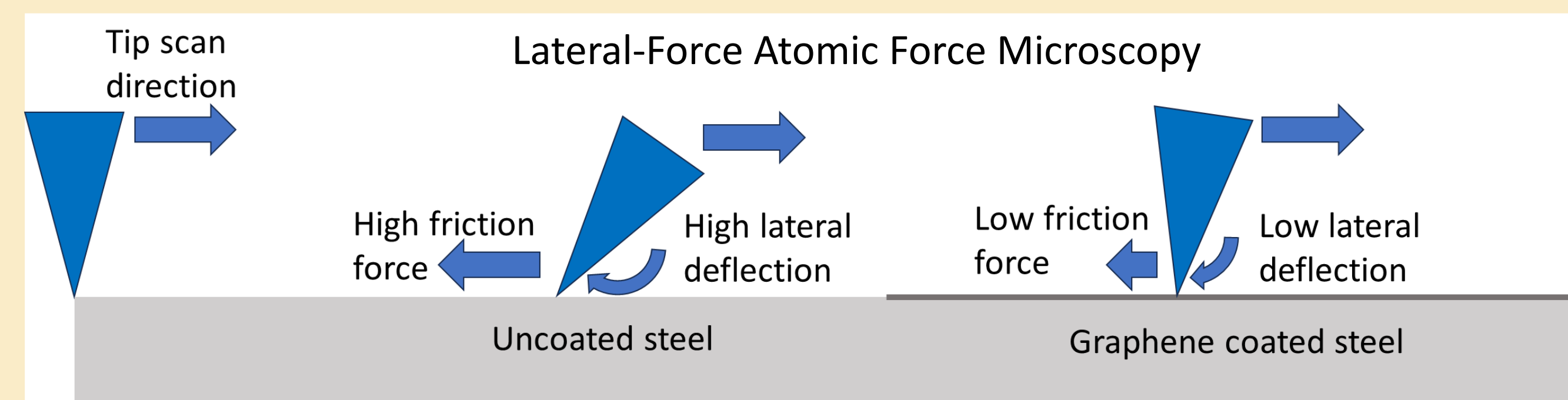


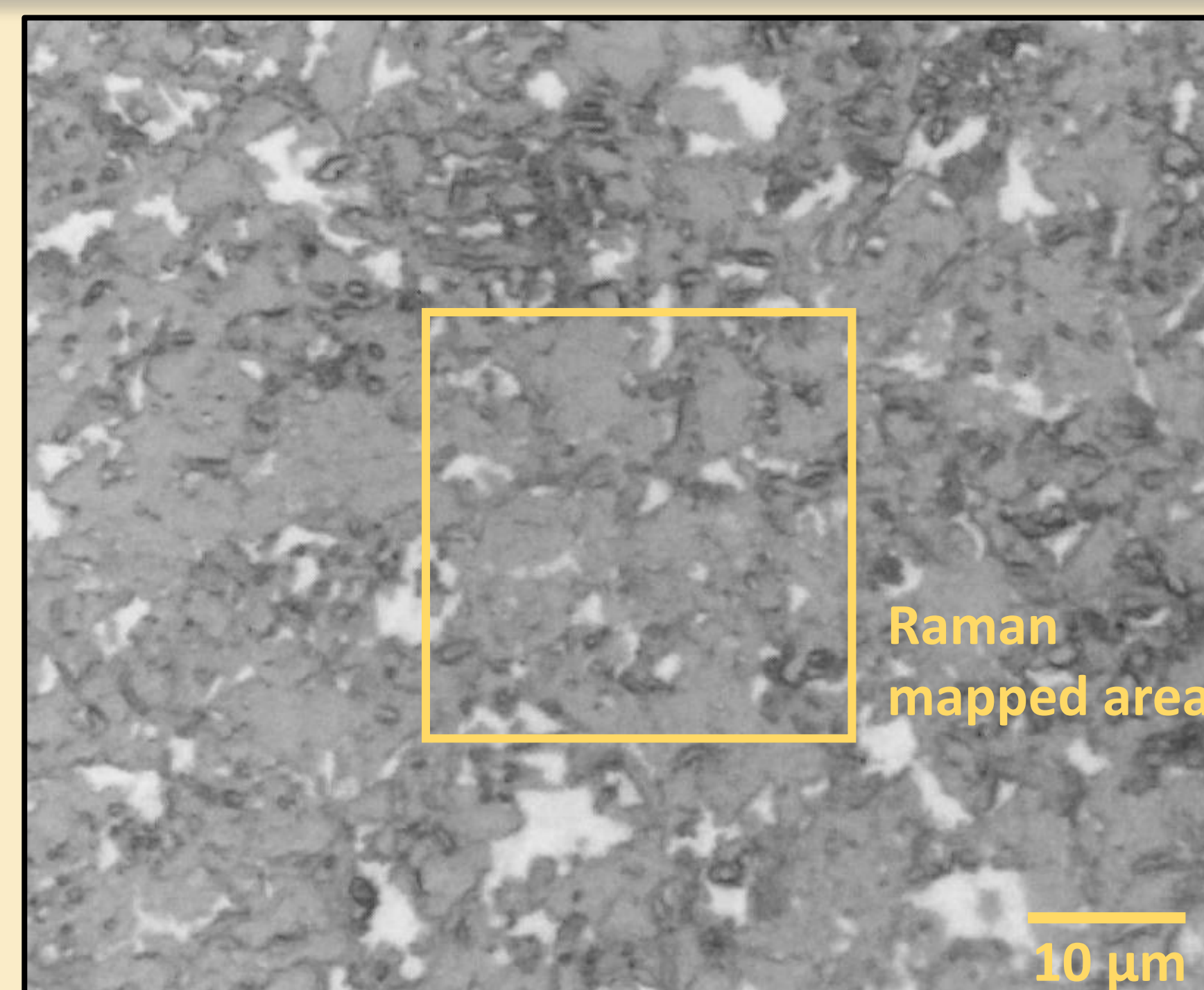
Introduction

Graphene is a promising nanoscale, low-friction coating for low-alloy steels[1]. Other studies of transferred graphene on steel showed that large graphene grain sizes[2] and increased thickness[3] reduce friction. This study examines how grain size and thickness of graphene grown by chemical vapor deposition (CVD) on 8620 low-alloy steel surfaces affect friction, using lateral-force atomic force microscopy (LF-AFM). This will lead to an understanding of how surface friction properties of the 8620 steel can be tailored with graphene coatings.

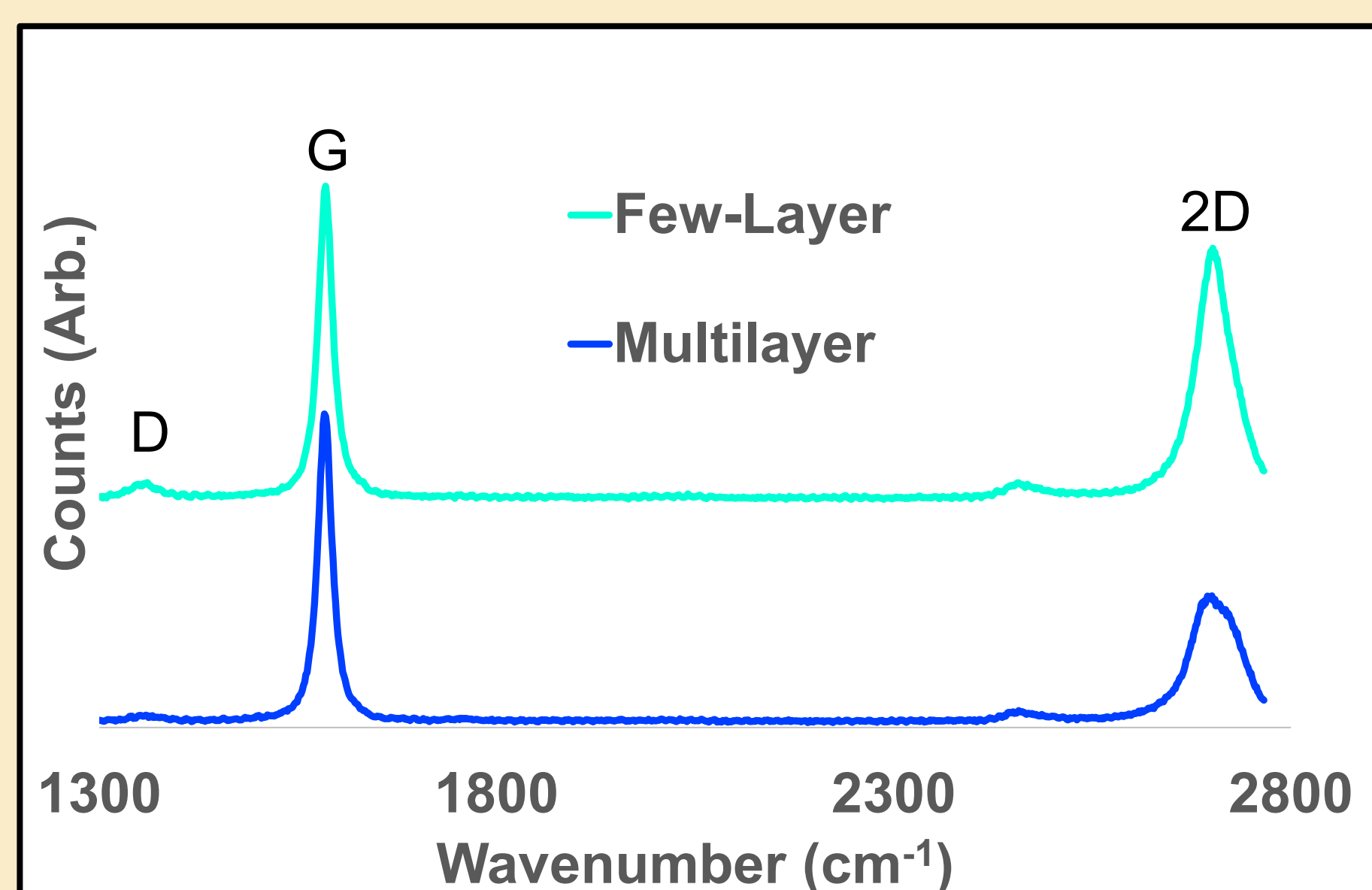
In LF-AFM, differences in tip-surface interactions cause the probe to twist as it slides on the specimen surface, measured as lateral deflection. Micrographs from LF-AFM highlight the surface properties such as friction force.



Microscopy and Raman characterization of graphene thickness



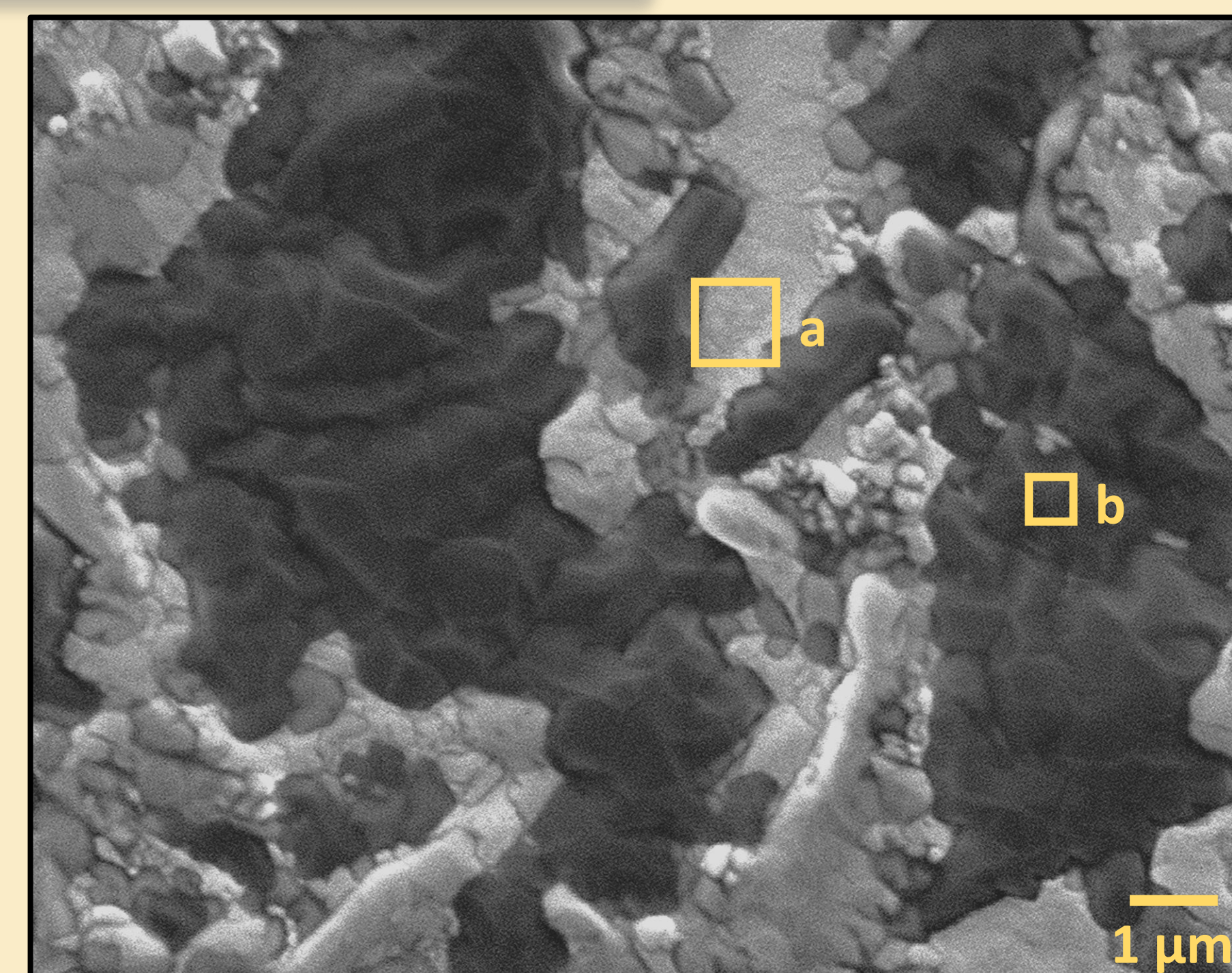
Optical microscopy was used to select large areas for Raman mapping. Dark areas are typically multilayer graphene (MLG) and light areas are few-layer graphene (FLG) or uncoated.



The relative thickness of 30 μm by 30 μm areas from the optical images were mapped by collecting Raman spectra point by point and analyzing their peaks.

The 2D:G peak ratio heatmap (right) shows MLG or FLG areas.

- Blue areas are primarily MLG with a low 2D:G peak ratio
- Teal and green areas are primarily FLG or single-layer graphene with a higher 2D:G ratio.

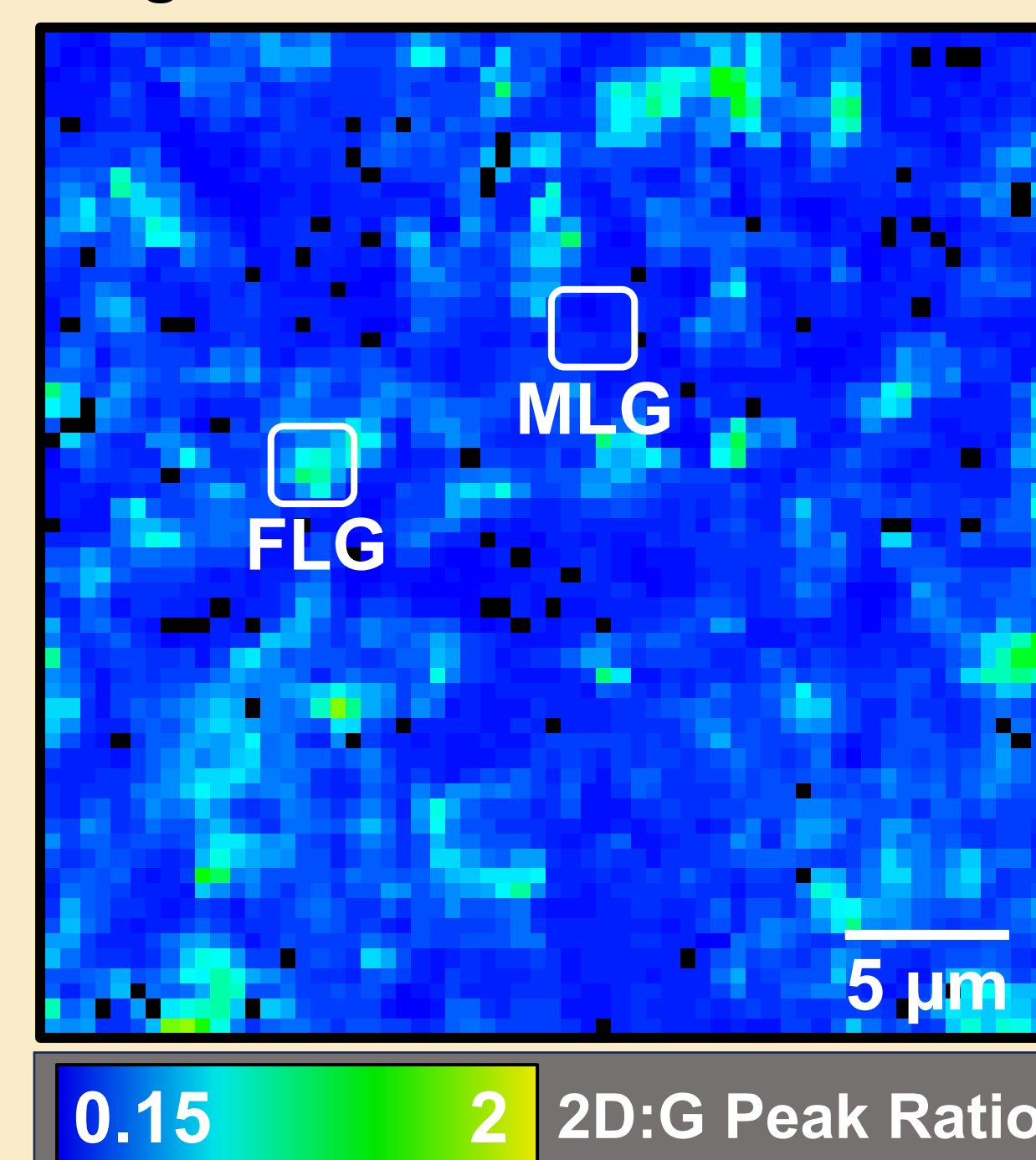


The SEM image shows a mixed FLG and uncoated steel area (box a) and example of an MLG area (box b). These and other similar areas were selected for LF-AFM.

Raman spectroscopy was used to characterize graphene thickness and grain size.

Selected spectra of FLG and MLG (left).

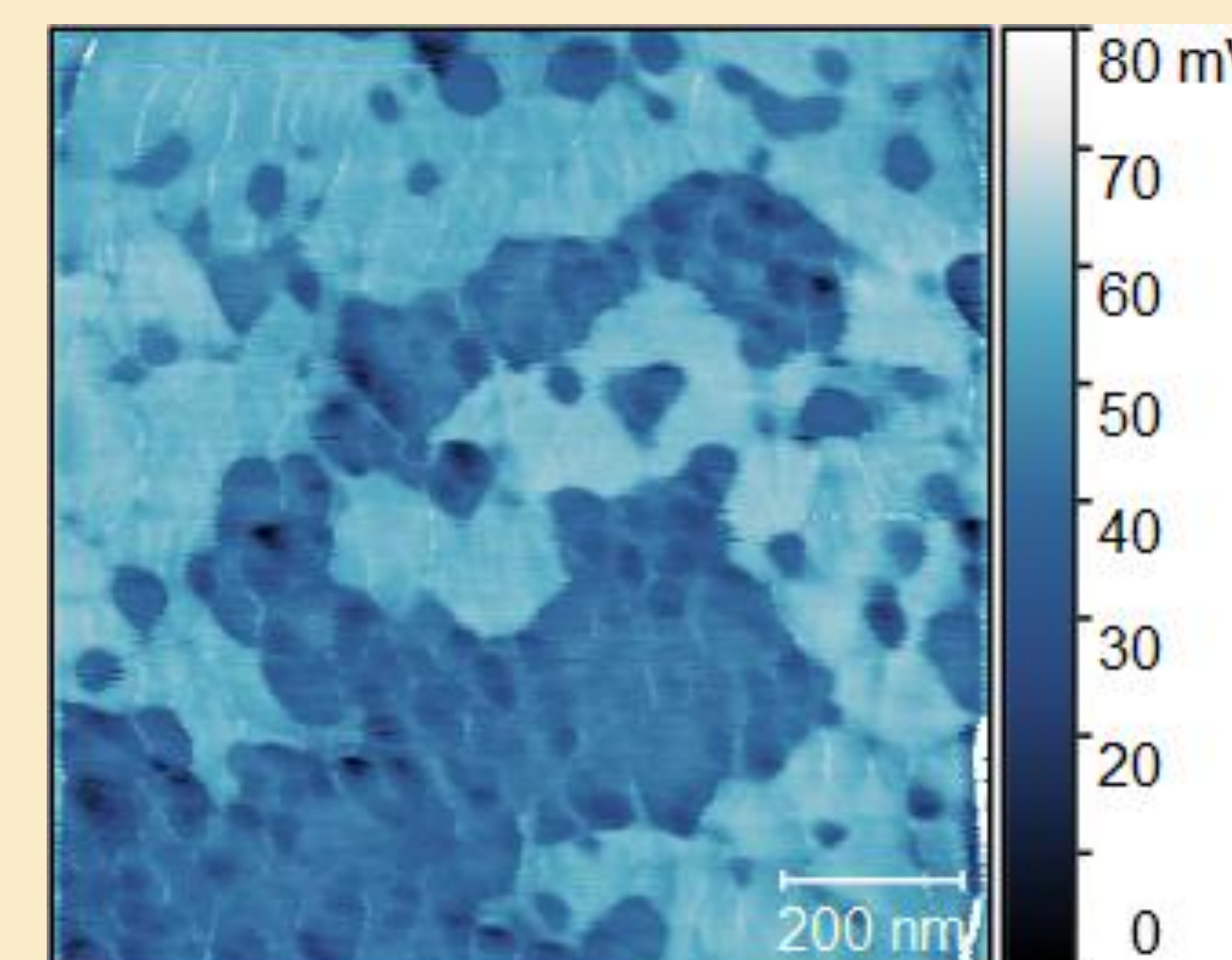
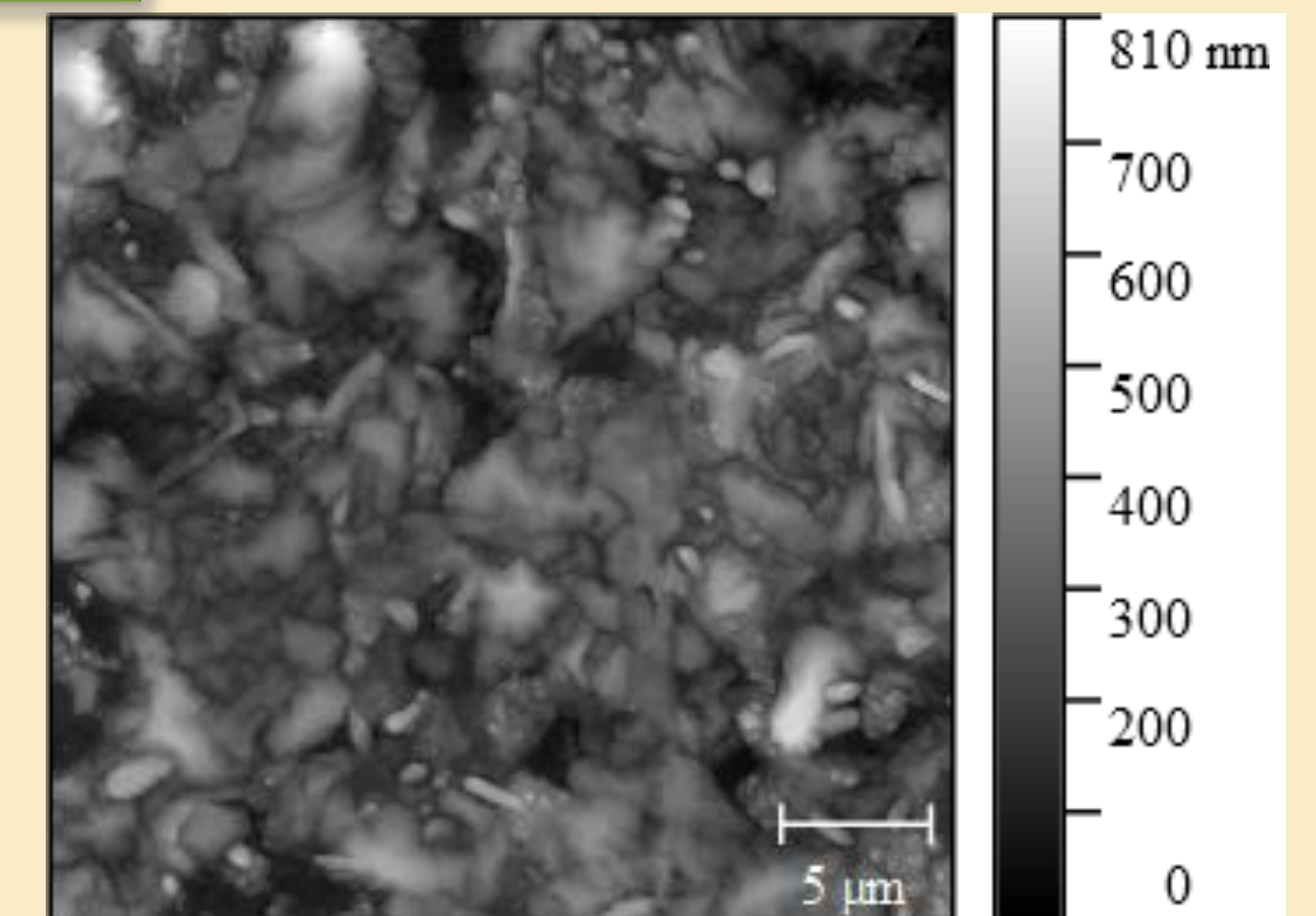
- The MLG (blue) has a low 2D peak (thicker graphene), and a low D peak (caused by defects such as grain boundaries).
- A FLG spectrum (teal) has a higher 2D peak (thinner graphene), and higher D peak (smaller grain size, more grain boundaries in the collection spot).



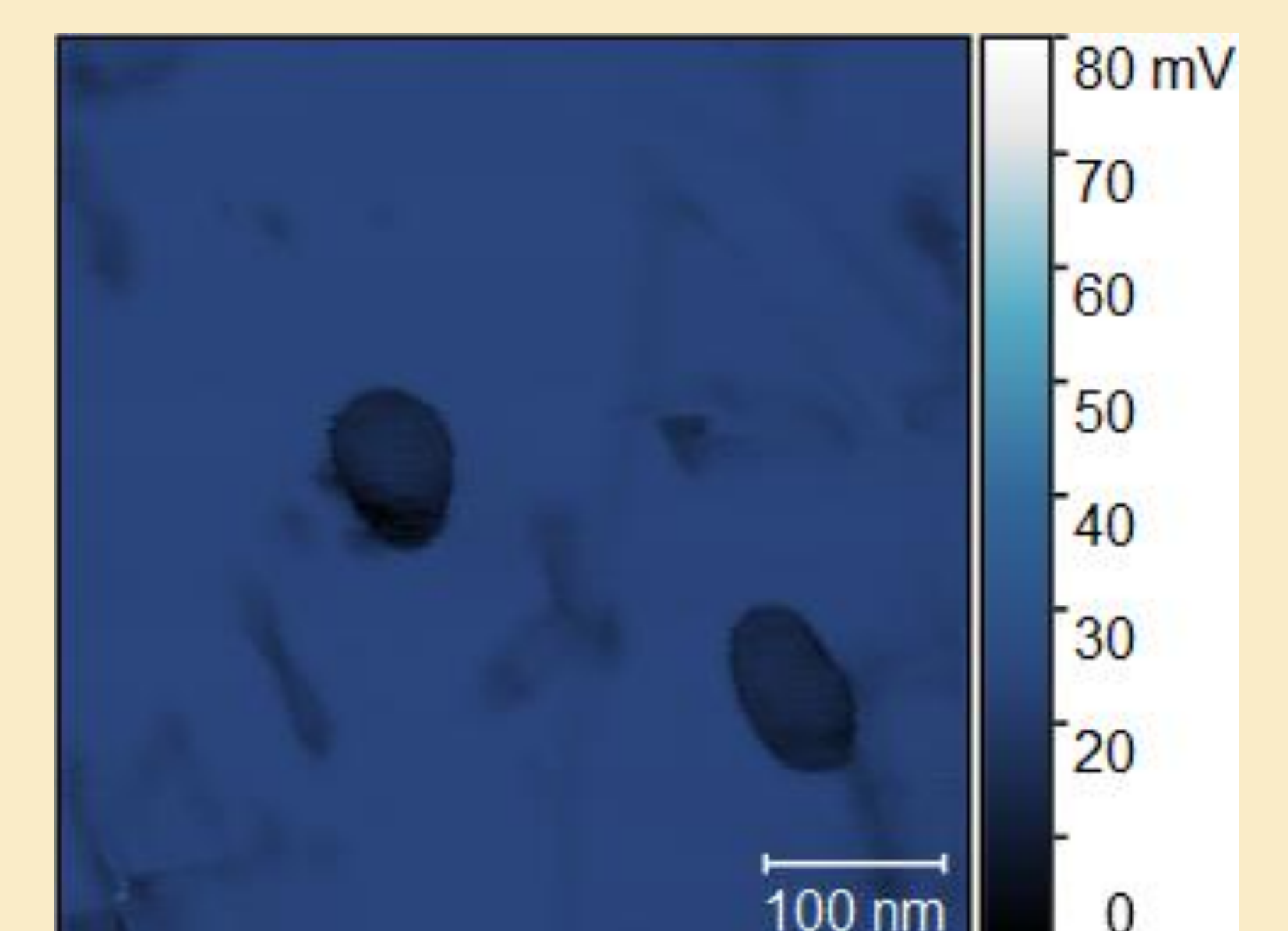
LF-AFM measurement of graphene friction force

AFM topography micrographs were collected from the same areas using a Nanosurf FlexAFM. By aligning optical images, Raman heatmaps, and AFM micrographs, 500 nm by 500 nm areas of MLG, FLG or uncoated areas were selected for LF-AFM analysis.

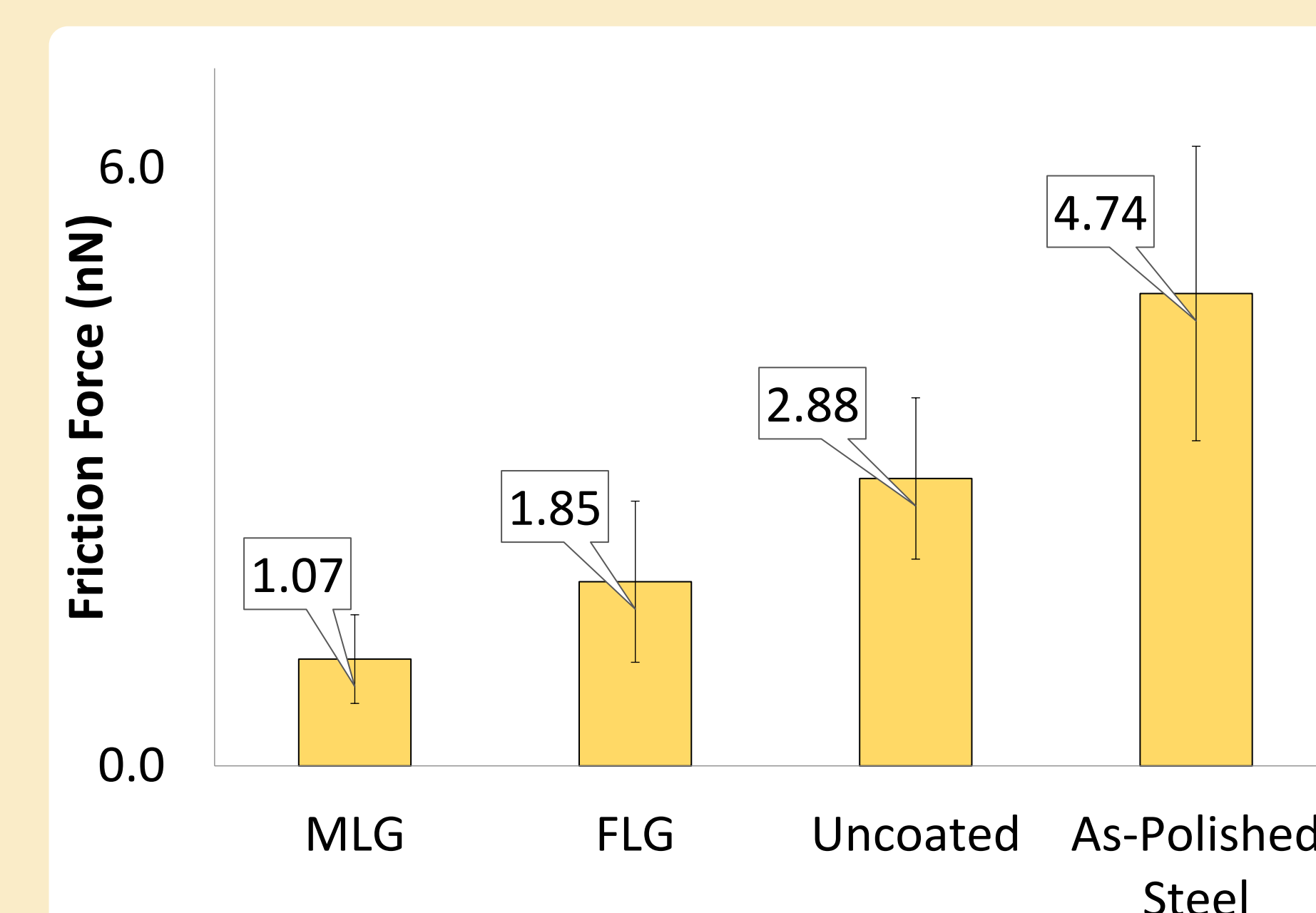
Lower lateral force (in mV) in LF-AFM micrographs (see below) corresponds to lower friction force, most apparent for MLG areas with low, uniform friction force. FLG areas have lower friction force than uncoated steel but are higher than MLG areas.



The LF-AFM micrograph comparing the edge of a FLG area with uncoated steel highlights the difference in friction force. The FLG has lower friction force, and some thicker areas match the low friction of MLG. This micrograph was acquired from box "a" in the SEM image.



The LF-AFM micrograph of the MLG with low overall friction force and good uniformity (no high friction areas caused by uncoated patches or defects) for thicker graphene. This micrograph was acquired from box "b" in the SEM image.



The average friction force for MLG, FLG, gaps in the coating, and polished steel with no graphene coating were determined from 40 micrographs (10 each). MLG areas have a 78% reduction in friction force compared to as-polished steel and FLG areas have a 61% improvement. The cantilever spring constant and lateral deflection sensitivity were calibrated in the Nanosurf software to estimate average friction force.

Conclusion

This study demonstrates that CVD graphene coatings reduce low-alloy steel friction, where graphene grain size and thickness play critical roles for friction reduction. Changing the CVD parameters for graphene growth to manipulate graphene thickness and grain size may enable tailored friction properties of the steel.

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